8.8 Improper Integrals

Up to now, we have considered definite integrals of the form $I = \int_a^b f(x)dx$, where the integrand f is **continuous** over the interval $[a, b]$. Such an integral I is necessarily a finite number and is called a ***proper integral***. In practice, we may encounter integrals for which the domain is infinite (infinite limits of integration) or the range of the integrand is infinite. In either case, the integrals are said to be ***improper*** and are calculated as limits.

Remark:
A definite integral $I = \int_a^b f(x)dx$ is

1. **proper** if f is continuous over the interval $[a, b]$;
2. **improper of** (i) Type I: if $a \to -\infty$ or $b \to \infty$. (Domain is infinite)
   (ii) Type II: if $f(x) \to \pm\infty$ at some points on $[a, b]$. (Range is infinite)

Example 1:

(1) $\int_{-1}^{1} \frac{1}{1+x^2}dx$ is a proper definite integral since $f(x) = \frac{1}{1+x^2}$ is continuous over the interval $[-1, 1]$.

(2) $\int_{1}^{\infty} \frac{1}{x(1+x^2)}dx$ is an improper definite integral of Type 1 since it is an integral with infinite limits of integration.

(3) $\int_{-1}^{1} \frac{1}{x(1+x^2)}dx$ is an improper definite integral of Type 2 since $0 \in [-1, 1]$ but $f(x) \to \infty$ as $x \to 0$.

Definition 1: (Improper Integrals of Type I)

Integrals with infinite limits of integration are ***improper integrals of Type I***.

1. If $f(x)$ is continuous on $[a, \infty)$, then $\int_a^{\infty} f(x)dx = \lim_{b\to\infty} \int_a^b f(x)dx$.
2. If $f(x)$ is continuous on $(-\infty, b]$, then $\int_{-\infty}^{b} f(x)dx = \lim_{a\to-\infty} \int_a^b f(x)dx$.
3. If $f(x)$ is continuous on $(-\infty, \infty)$, then $\int_{-\infty}^{\infty} f(x)dx = \lim_{a\to-\infty} \int_a^c f(x)dx + \lim_{b\to\infty} \int_c^b f(x)dx$. where c is any real number.

In each case, if the limit exists and is finite, we say that the improper integral ***converges*** and that the limit is the ***value*** of the improper integral. If the limit fails to exist, the improper integral ***diverges***.

The choice of c in Part 3 of the definition is unimportant. We can evaluate or determine the convergence or divergence of the improper integral with any convenient choice of c.

Remark: If $y = f(x)$ is nonnegative and integrable over $[a, b]$, then the area under the curve $y = f(x)$ over $[a, b]$ is given by $\int_a^b f(x)dx$. An improper integral (Type I or Type II) also can be interpreted as an area if $f \geq 0$ on the interval of integration. The area is infinite if the improper integral diverges.

Example 2:

Is the area under the curve $y = \frac{\ln x}{x^2}$ for $x \geq 1$ finite? If so, what is its value?

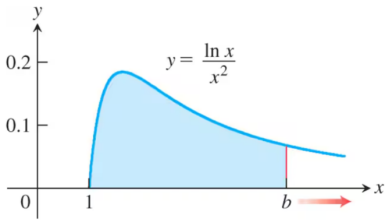
Solution:

Since $f(x) = \frac{\ln x}{x^2}$ is continuous on $[1, \infty)$, then $\int_1^\infty \frac{\ln x}{x^2}\,dx = \lim_{b\to\infty} \int_1^b \frac{\ln x}{x^2}\,dx$.

To evaluate $\int \frac{\ln x}{x^2}\,dx$, we use integration by parts with $u = \ln x$ and $dv = \frac{dx}{x^2}$.

Then, $du = \frac{dx}{x}$ and $v = -\frac{1}{x}$.

$$\int \frac{\ln x}{x^2}\,dx = \int u\,dv = uv - \int v\,du = -\frac{\ln x}{x} - \int \frac{1}{x}\left(-\frac{1}{x}\right)dx = +C.$$



$$\int_1^\infty \frac{\ln x}{x^2}\,dx = \lim_{b\to\infty} \int_1^b \frac{\ln x}{x^2}\,dx = \lim_{b\to\infty} \left[-\left(\frac{\ln x}{x} + \frac{1}{x}\right)\right]_1^b$$

$$= \lim_{b\to\infty} \left[-\left(\frac{\ln b}{b} + \frac{1}{b}\right) + (0+1)\right] = \lim_{b\to\infty} \left[-\frac{\frac{1}{b}}{1} - 0 + 1\right]$$

l'Hôpital's Rule

$$= 0 + 1 = 1.$$

Thus, the improper integral converges to 1.

Since $f(x) = \frac{\ln x}{x^2} \geq 0$ for $x \geq 1$, the improper integral can be interpreted as the (finite) area $y = \frac{\ln x}{x^2}$ for $x \geq 1$ and the area has finite value 1.

Example 3: Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$.

Solution:

According to Part 3 of the definition, we can choose $c = 0$ and write

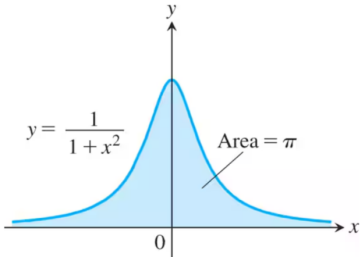
$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_0^{\infty} \frac{1}{1+x^2}\,dx.$$

$$\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx = \lim_{a\to-\infty} \int_a^0 \frac{1}{1+x^2}\,dx = \lim_{a\to-\infty} \tan^{-1}x\,]_a^0$$

$$= \lim_{a\to-\infty} (\tan^{-1}0 - \tan^{-1}a) = 0 - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2}.$$

$$\int_0^{\infty} \frac{1}{1+x^2}\,dx = \lim_{b\to\infty} \int_0^b \frac{1}{1+x^2}\,dx = \lim_{b\to\infty} \tan^{-1}x\,]_0^b$$

$$= \lim_{b\to\infty} (\tan^{-1}b - \tan^{-1}0) = \frac{\pi}{2} - 0 = \frac{\pi}{2}.$$



Thus, $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi$.

Since $f(x) = \frac{1}{1+x^2} > 0$, the improper integral can be interpreted as the (finite) area beneath the curve and above the x-axis.

The integral $\int_1^\infty \frac{1}{x^p}dx$ **:**

$$\int_1^\infty \frac{1}{x^p}dx = \begin{cases} \text{converges to } \frac{1}{p-1} & \text{if } p>1 \\ \text{diverges} & \text{if } p \le 1 \end{cases}.$$

The integral $\int_1^\infty \frac{1}{x^p}dx$ is useful in testing some integrals for convergence.

Example 4:

For what values of p does the integral $\int_1^\infty \frac{1}{x^p}dx$ converge? When the integral does converge, what is its value?

Solution:

If $p \neq 1$, $\int_1^b \frac{dx}{x^p} = \frac{x^{-p+1}}{-p+1}\Big]_1^b = \frac{1}{1-p}(b^{-p+1}-1) = \frac{1}{1-p}\left(\frac{1}{b^{p-1}}-1\right)$.

Thus,

$$\int_1^\infty \frac{dx}{x^p} = \lim_{b\to\infty}\int_1^b \frac{dx}{x^p}$$

$$= \lim_{b\to\infty}\left[\frac{1}{1-p}\left(\frac{1}{b^{p-1}}-1\right)\right] = \begin{cases} \frac{1}{p-1}, & p>1 \\ \infty, & p<1 \end{cases}$$

Constant

because

$$\lim_{b\to\infty}\frac{1}{b^{p-1}} = \begin{cases} 0, & p>1 \\ \infty, & p<1. \end{cases}$$

Therefore, the integral converges to the value $1/(p-1)$ if $p>1$ and it diverges if $p<1$.

If $p=1$, the integral also diverges:

$$\int_1^\infty \frac{dx}{x^p} = \int_1^\infty \frac{dx}{x}$$

$$= \lim_{b\to\infty}\int_1^b \frac{dx}{x}$$

$$= \lim_{b\to\infty} \ln x\Big]_1^b$$

$$= \lim_{b\to\infty}(\ln b - \ln 1) = \infty.$$

Example 5: Which of the following integrals are convergent and which divergent?

(1) $\int_1^\infty \frac{1}{\sqrt{x}}dx$ (2) $\int_1^\infty \frac{1}{x^{3/2}}dx$

(3) $\int_1^\infty x^3 dx$ (4) $\int_1^\infty x^{-4/3}dx$

(5) $\int_1^\infty \sqrt{x}dx$

Definition 2: (Improper Integrals of Type II)

Integrals of functions that become infinite at a point within the interval of integration are **improper integrals of Type II**.

1. If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then $\int_a^b f(x)dx = \lim_{c\to a^+} \int_c^b f(x)dx$.

2. If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then $\int_a^b f(x)dx = \lim_{c\to b^-} \int_a^c f(x)dx$.

3. If $f(x)$ is discontinuous at c where $a < c < b$, and continuous $[a, c) \cup (c, b]$ then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

In each case, if the limit exists and is finite, we say that the improper integral **converges** and that the limit is the **value** of the improper integral. If the limit does not exist, the integral **diverges**.

In Part 3 of the definition, the integral on the left side of the equation converges if *both* integrals on the right side converge; otherwise, it diverges.

Example 6: Investigate the convergence of $\int_0^1 \frac{1}{1-x}dx$.

Solution:

The integrand $f(x) = \frac{1}{1-x}$ is continuous on $[0, 1)$ but is discontinuous at $x = 1$ and becomes infinite as $x \to 1^-$. We evaluate the integral as

$$\lim_{b\to 1^-} \int_0^b \frac{1}{1-x}dx = \lim_{b\to 1^-} \left[-\ln|1-x|\right]_0^b$$
$$= \lim_{b\to 1^-} [-\ln(1-b) + 0] = \infty.$$

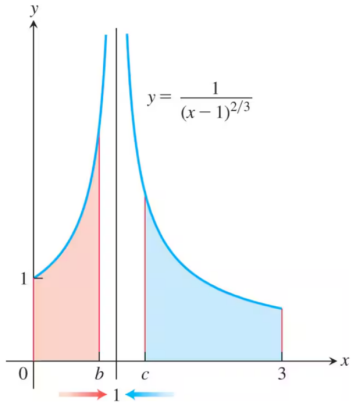
The limit is infinite, so the integral diverges.

Example 7:

Evaluate $\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

Solution:

The integrand has a vertical asymptote at $x = 1$ and is continuous on [0, 1) and (1, 3]. Thus, by Part 3 of the definition above,



$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = \int_0^1 \frac{dx}{(x-1)^{2/3}} + \int_1^3 \frac{dx}{(x-1)^{2/3}}.$$

$$\int_0^1 \frac{dx}{(x-1)^{2/3}} = \lim_{b\to 1^-} \int_0^b \frac{dx}{(x-1)^{2/3}}$$

$$= \lim_{b\to 1^-} 3(x-1)^{1/3}\Big]_0^b$$

$$= \lim_{b\to 1^-} \left[3(b-1)^{1/3} + 3\right] = 3$$

$$\int_1^3 \frac{dx}{(x-1)^{2/3}} = \lim_{c\to 1^+} \int_c^3 \frac{dx}{(x-1)^{2/3}}$$

$$= \lim_{c\to 1^+} 3(x-1)^{1/3}\Big]_c^3$$

$$= \lim_{c\to 1^+} \left[3(3-1)^{1/3} - 3(c-1)^{1/3}\right] = 3\sqrt[3]{2}$$

Therefore, $\int_0^3 \frac{dx}{(x-1)^{2/3}} = 3 + 3\sqrt[3]{2}.$

Tests for Convergence and Divergence

When an improper integral can't be evaluated by the Fundamental Theorem of Calculus because an antiderivative can't be found, we may still able to determine whether the integral converges (or diverges) by comparing it with one integral whose convergence (divergence) properties are known.

Theorem 1: Direct Comparison Test

Let f and g be continuous on $[a, \infty)$. Then

(1) $0 \le f(x) \le g(x)$ for all $x \ge a$ and $\int_a^\infty g(x)dx$ converges $\Rightarrow \int_a^\infty f(x)dx$ converges.

(2) $0 \le g(x) \le f(x)$ for all $x \ge a$ and $\int_a^\infty g(x)dx$ diverges $\Rightarrow \int_a^\infty f(x)dx$ diverges.

CAUTION: **DO NOT USE** the Direct Comparison Test to test $\int_a^\infty f(x)dx$ when

(i) $f(x) \le g(x)$ and $\int_a^\infty g(x)dx$ diverges; or (ii) $g(x) \le f(x)$ and $\int_a^\infty g(x)dx$ converges.

Example 8:

(a) For $x \ge 1$, $0 \le \sin^2 x \le 1 \Rightarrow 0 \le \dfrac{\sin^2 x}{x^2} \le \dfrac{1}{x^2}$.

Furthermore, $\int_1^\infty \dfrac{1}{x^2}dx$ converges.

$\therefore$ By the direct comparison test, $\int_1^\infty \dfrac{\sin^2 x}{x^2}dx$ converges.

(b) For $x \ge 1$, $0 \le \dfrac{1}{\sqrt{x^2}} \le \dfrac{1}{\sqrt{x^2 - 0.1}} \Rightarrow 0 \le \dfrac{1}{x} \le \dfrac{1}{\sqrt{x^2 - 0.1}}$.

Furthermore, $\int_1^\infty \dfrac{1}{x}dx$ diverges.

$\therefore$ By the direct comparison test, $\int_1^\infty \dfrac{dx}{\sqrt{x^2 - 0.1}}$ diverges.

Theorem 3: Limit Comparison Test

If the positive functions f and g are continuous on $[a, \infty)$ and if $\lim_{x\to\infty} \dfrac{f(x)}{g(x)} = L$ is a positive finite limit, then $\int_a^\infty f(x)dx$ and $\int_a^\infty g(x)dx$ both converge or both diverge.

Example 9:

Show that $\int_1^\infty \frac{dx}{1+x^2}$ converges by comparison with $\int_1^\infty \frac{dx}{x^2}$. Find and compare the two integrals values.

Solution:

The functions $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{1+x^2}$ are positive and continuous on [1, ∞]. Also,

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{1/x^2}{1/(1+x^2)} = \lim_{x\to\infty} \frac{1+x^2}{x^2} = \lim_{x\to\infty} \left(\frac{1}{x^2} + 1\right) = 0 + 1 = 1,$$

a positive limit. Therefore, $\int_1^\infty \frac{dx}{1+x^2}$ converges because $\int_1^\infty \frac{dx}{x^2}$ converges.

However, $\int_1^\infty \frac{dx}{x^2} = 1$ and $\int_1^\infty \frac{dx}{1+x^2} = \frac{\pi}{4}$. (Exercise)

Example 10: Test $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ for convergence.

Solution:

The integrand suggests a comparison of $f(x)=\frac{1-e^{-x}}{x}$ with $g(x)=\frac{1}{x}$. However, we cannot use the Direct Comparison Test because $f(x) \le g(x)$ and $\int_1^\infty \frac{1}{x}\,dx$ diverges.

On the other hand, using the Limit Comparison Test, we find that

$$\lim_{x\to\infty}\frac{f(x)}{g(x)}=\lim_{x\to\infty}\frac{(1-e^{-x})/x}{1/x}=\lim_{x\to\infty}\left(1-\frac{1}{e^x}\right)=1 \text{ is a positive finite limit.}$$

$\therefore$ $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ diverges because $\int_1^\infty \frac{1}{x}\,dx$ diverges.

Example 11:
Test the following integrals for convergence.

(1) $\int_0^{\ln 2} \frac{1}{x^2 e^{1/x}}\,dx$ (2) $\int_1^\infty \frac{dx}{\sqrt{x^6-1}}$ (3) $\int_\pi^\infty \frac{2+\cos x}{x}\,dx$

(Remark: To test an improper integral for convergence, we may (i) integrate directly, (ii) use the Direct Comparison Test, or (iii) use the Limit Comparison Test.)

Solution:

(1) Let $u=\frac{1}{x}$. Then $du=-\frac{1}{x^2}\,dx$. $\int \frac{1}{x^2 e^{1/x}}\,dx=-\int \frac{du}{e^u}=-(-e^{-u})+C=e^{-1/x}+C$.

$$\int_0^{\ln 2} \frac{1}{x^2 e^{1/x}}\,dx=\lim_{a\to 0^+}\int_a^{\ln 2} \frac{1}{x^2 e^{1/x}}\,dx=\lim_{a\to 0^+}\left[e^{-\frac{1}{x}}\right]_a^{\ln 2}=\lim_{a\to 0^+}\left[e^{-\frac{1}{\ln 2}}-e^{-\frac{1}{a}}\right]=e^{-\frac{1}{\ln 2}}.$$

$\therefore$ $\int_0^{\ln 2} \frac{1}{x^2 e^{1/x}}\,dx$ converges.

(2) Let $f(x)=\frac{1}{\sqrt{x^6-1}}$ and with $g(x)=\frac{1}{x^3}$.

$\lim_{x\to\infty}\frac{f(x)}{g(x)}=\lim_{x\to\infty}\frac{1/\sqrt{x^6-1}}{1/x^3}=\lim_{x\to\infty}\frac{x^3}{\sqrt{x^6-1}}=\lim_{x\to\infty}\frac{x^3}{x^3\sqrt{1+(1/x^6)}}=1$ is a positive finite limit.

By the limit comparison test, $\int_1^\infty \frac{dx}{\sqrt{x^6-1}}$ converges because $\int_1^\infty \frac{1}{x^3}\,dx$ converges.

(3) For $x \ge 1$, $1 \le 2+\cos x \le 3 \Rightarrow 0<\frac{1}{x}\le\frac{2+\cos x}{x}$.

Furthermore, $\int_1^\infty \frac{1}{x}\,dx$ diverges.

$\therefore$ By the direct comparison test, $\int_1^\infty \frac{2+\cos x}{x}\,dx$ diverges.